



Module Presenter's Manual

for

Data Processing with XML and JSON

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Ver. 1.0***

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	December 2023	New	-

Table of Contents

Sr. No.	Details	Page No.
1.	Introduction	1
2.	Information on Session Allocation	2
3.	Module Deliverables on OnlineVarsity	2
4.	Week-wise Session Schedule	3
5.	Session Coverage	4
6.	Library References	8

1. Introduction

At the end of this course, students will be able to:

- Interpret the world of data interchange and processing through XML and JSON
- Discover how to use XML, syntax, attributes, and schema
- Explain about DTDs
- Learn XSLT and XPath expressions
- Describe JSON and data types of JSON
- Create and parse JSON messages with JavaScript
- Learn advanced JSON features such as Web APIs, JSON HTTP and Files, and Data Storage using JSON
- Learn about Cross-Origin Resource Sharing (CORS)
- Describe different CORS origins and types
- Explain the kind of CORS requests sent by a browser

2. Information on Session Allocation

Module	Online (No. of Hours)	Self-Study (No. of Hours)
Data Processing with XML and JSON	12	8

Throughout this Presenter's Manual, the module **Data Processing with XML and JSON** will be referred to as **XML-JSON**.

3. Module Deliverables available on OnlineVarsity

To aid the learning process, following are the deliverables:

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Week-wise Session Schedule

- A Session is of 2 hours duration.

➤ Week-Wise Schedule

Week	Day 1	Day 2		Day 3		Day 4
1	Session 1 XML- JSON – TL1	Session 2 XML- JSON – TL2		Session 3 XML-JSON – TL3	S1 & S2	Session 4 XML-JSON – TL4
2	Session 5 XML - JSON– TL5	Session 6 XML- JSON – TL6	S3 & S4			

TL: Online Session

S: Self Study (over and above the module duration)

4. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	XML-JSON – TL1	All the topics as listed below from Session 1 and Session 2 of <i>Data Processing with XML and JSON</i> book should be covered in this session. <u>Session 1 – XML and XML Validations</u> ➤ Define XML and explain its features ➤ Illustrate XML Attributes, Comments, Trees, and Validations ➤ Explain XML DTD with its types and describe working with DTDs <u>Session 2 – XML Parser and DOM</u> ➤ Describe XML Parsers ➤ Outline XML Document Object Model (DOM) and list various types of XML Database ➤ Explain XML Namespaces	<u>Data Processing with XML and JSON</u> SG - Session 1 & 2 XP - Session 1 & 2 TG - Session 1 & 2

Session No.	Session Title	Session Details	Deliverables' Mapping
2	XML-JSON – TL2	All the topics as listed below from Session 3 and Session 4 of <i>Data Processing with XML and JSON</i> book should be covered in this session. <u>Session 3 – XPath</u> ➤ Explain XPath ➤ Learn XPath Nodes ➤ Describe XPath Operators <u>Session 4 – XSLT</u> ➤ Describe XSLT and its various characteristics ➤ Explain the XSLT syntax ➤ List different XSLT Elements	<u>Data Processing with XML and JSON</u> SG - Session 3 & 4 XP - Session 3 & 4 TG - Session 3 & 4
3	XML-JSON – TL3	The Try It Yourself questions of Session 1 to Session 4 of <i>Data Processing with XML and JSON</i> book should be covered in this session.	<u>Data Processing with XML and JSON</u> Session 1-4
	S1 & S2	All the Assignments from Session 1 to Session 4 of <i>Data Processing with XML and JSON</i> book should be covered in this session.	<u>Data Processing with XML and JSON (OnlineVarsity)</u> Session 1-4

Session No.	Session Title	Session Details	Deliverables' Mapping
4	XML-JSON – TL4	All the topics as listed below from Session 5 and Session 6 of <i>Data Processing with XML and JSON</i> book should be covered in this session. <u>Session 5 – JSON</u> ➤ Describe JSON and explain its various features ➤ List the differences between JSON and XML ➤ Explain the JSON object and array ➤ Explain nested JSON <u>Session 6 – Working with JSON</u> ➤ Explain various data types used in JSON ➤ Explain the fundamental data structures available in JSON ➤ Define JSON schema and explain its role in defining data structure constraints ➤ Explain how to create and parse JSON Messages with JavaScript	<u>Data Processing with XML and JSON</u> SG - Session 5 & 6 XP - Session 5 & 6 TG - Session 5 & 6

Session No.	Session Title	Session Details	Deliverables' Mapping
5	XML-JSON – TL5	All the topics as listed below from Session 7 and Session 8 of <i>Data Processing with XML and JSON</i> book should be covered in this session. <u>Session 7 – Advanced JSON Features</u> ➤ Identify JSON's support with programming languages ➤ Define the usage of JSON in Web APIs and MIME types ➤ Describe JSON's role in data storage, security, and data portability <u>Session 8 – Cross-Origin Resource Sharing (CORS)</u> ➤ Elaborate CORS and its terminology ➤ Summarize the benefits of implementing CORS, Origin and explore its different types ➤ Outline the process of implementing the CORS policy in Web browsers and various types of CORS requests sent by a Web browser	<u>Data Processing with XML and JSON</u> SG - Session 7 & 8 XP - Session 7 & 8 TG - Session 7 & 8
6	XML-JSON – TL6	The Try It Yourself questions of Session 5 to Session 8 of <i>Data Processing with XML and JSON</i> book should be covered in this session.	<u>Data Processing with XML and JSON</u> Session 5-8
	S3 & S4	All the Assignments from Session 5 to Session 8 of <i>Data Processing with XML and JSON</i> book should be covered in this session.	<u>Data Processing with XML and JSON (OnlineVarsity)</u> Session 5-8

6. Library References

➤	XML: THE COMPLETE REFERENCE by Heather Williamson
➤	Json for Beginners: Your Guide to Easily Learn Json in 7 Days by ICode Academy
➤	SCHAUM'S EASY OUTLINE XML by Ed Tittel

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